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REVIEW

Polypropylene foaming using supercritical carbon dioxide: a review on fundamentals, technology, and applications

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Abstract
Polymer foams are extensively employed in automotive, flame retardant, and food packaging owing to their lightweight, excellent insulation, high strength, thermal stability, and impact resistance. Polypropylene (PP) is regarded as an alternative to other thermoplastic polymers because of its low-cost, excellent heat resistance, and corrosion resistance. However, the inherent limitation stemming from the weak melt strength of linear PP, coupled with its low melt elasticity, presents significant challenges in the production of PP foams characterized by microsized and uniform cell distribution. To enhance the performance of PP foam, various modification methods such as long-chain branching, cross-linking, blending, and adding fillers are employed. This paper first provides an overview of the foaming process of supercritical carbon dioxide and analyzes the influence of chemical structure and crystal structure on PP foaming. Subsequently, methods to enhance PP foaming properties, such as branching, cross-linking, blending, and filler incorporation, are discussed. Various foaming technologies for PP, including batch foaming, extrusion foaming, and injection foaming, are delineated. Furthermore, the diverse applications of PP foams, encompassing automotive components, thermal insulation, flame retardancy, electromagnetic shielding, oil–water separation, and food packaging were outlined. Finally, the current research status of PP foams is summarized, and the prospects for high-performance PP are addressed.

Keywords Polypropylene · Foam · Cell structure · Foaming technology · Application

1 Introduction

Polymer foam is a crucial polymer material, often referred to as a porous polymer due to the abundance of minuscule cell holes within its matrix [1–7]. In comparison to unfoamed polymers, polymer foams exhibit numerous advantages, including low density, excellent thermal insulation, effective sound insulation, high specific strength, and corrosion resistance [8–16]. Owing to these advantages, polymer foams find extensive applications in diverse fields, including transportation, military, sports equipment, daily necessities, clothing, aerospace, construction, electrical, and electronic applications [17–24].

Polypropylene is a linear polymer with a polyhedric character, classified into three stereospecific configurations: isotactic, syndiotactic, and atactic [25]. Isotropic PP is extensively employed due to its favorable structural regularity, high crystallinity, elevated melting point, increased hardness and stiffness, and commendable mechanical properties. In comparison to the common polymers such as polystyrene (PS) and polyethylene (PE), PP offers a notably higher service temperature, reaching up to 130 °C, while the maximum service temperature for PS and PE is confined to 80 °C. PP also demonstrates greater rigidity compared to PE, with a flexural modulus of approximately 1.52 GPa, significantly surpassing PE's flexural modulus of 207 MPa. Furthermore, PP is inherently non-toxic and odorless and exhibits excellent corrosion resistance [26–30]. Moreover, PP emerges as an ideal alternative to other thermoplastic

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foams due to its elevated tensile modulus, low density, and cost-effectiveness.

Nevertheless, as a semi-crystalline polymer, PP exhibits high degree of crystallinity and low melt strength, contributing to the suboptimal quality of PP foams. Firstly, the weak melt strength results in cell walls lacking the resilience necessary to withstand the bi-directional tensile force from cell expansion. This leads to cell rupture, coalescence, and gas escape, consequently impacting both cell structure and foam performance. Further, CO_2 encounters difficulty in diffusing or dissolving into the crystalline area of PP, unlike the amorphous region where there are no such restrictions imposed by crystallization. During the cell nucleation stage, preferential nucleation occurs at the interface between the crystalline and amorphous regions, leading to an increased cell density. The presence of crystalline regions during the cell growth stage hampers cell growth, resulting in diminished cell homogeneity. This intricate interplay ultimately results in the formation of polymer foams characterized by uneven cell size and a non-uniform distribution of cell, thereby impacting foam performance. Therefore, several methods have been employed to improve PP foaming properties [31–35], including branching, crosslinking, blending, and adding fillers [36–39].

The conventional approach for preparing polymer foams employs chemical foaming technology [40]. However, the harmful gases released by chemical foaming agents can contribute to environmental pollution, and residual foaming agents may pose risks to human health. In addition, foams produced through chemical foaming typically exhibit larger cell sizes (more than 100 μm) and low cell density (less than 10^9 cells/cm^3). In contrast, supercritical foaming utilizes non-toxic and harmless gases, primarily supercritical carbon dioxide ($ScCO_2$) and supercritical nitrogen (ScN_2), as foaming agents [41]. Foams produced via supercritical technology can control cell sizes at the micron or even nanometer scale, with cell density reaching as high as 10^9–10^{15} cells/cm^3 [42–44]. Furthermore, foams by supercritical technology exhibit greater impact strength, improved thermal stability, toughness, good acoustic performance, and lower thermal conductivity. Therefore, using $ScCO_2$ as a blowing agent for preparing PP foams holds significant market potential and environmental value.

This paper reviews the research progress of PP foaming using supercritical carbon dioxide. Firstly, the principles of the foaming process with supercritical carbon dioxide was introduced, comprising the formation of a homogeneous system, cell nucleation, cell growth, and cell stereotyping. Subsequently, the methods to improve PP foaming properties were discussed, including blending, branching, crosslinking, and the addition of fillers. Further, three PP foaming technologies were presented, including batch foaming, extrusion foaming, and injection foaming. Subsequently, the applications of PP foam in various fields such as automotive, flame retardancy, conductivity, electromagnetic shielding, oil–water separation, and food packaging were discussed. Lastly, the paper summarizes the challenges and limitations observed in PP foaming processes in recent years.

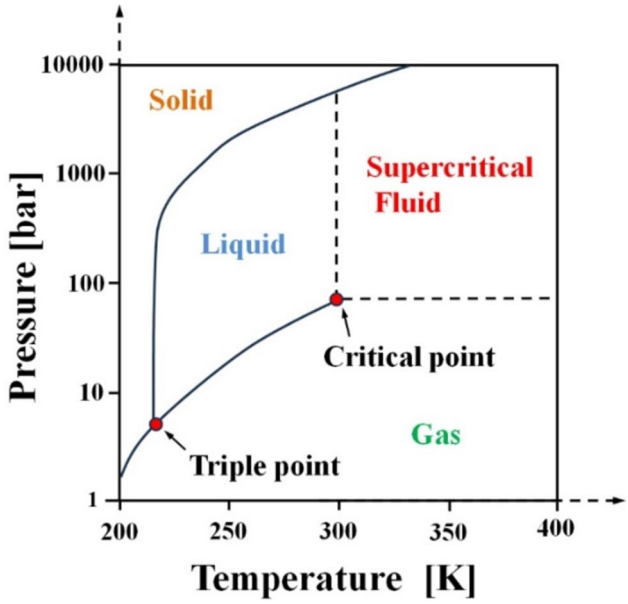

Fig. 1 Supercritical CO_2 P–T diagram

Table 1 Value ranges of density, viscosity, and diffusivity for gas, supercritical fluid, and liquid

Properties	Gas	Supercritical fluid	Liquid
Density (g/L)	0.6–2	100–1000	600–1600
Viscosity (μPa.s)	10–50	10–30	200–3000
Diffusivity × 10^{-9} (m^2/s)	10,000	10–100	0.2–2

2 Foaming mechanisms of PP

2.1 Supercritical CO_2 foaming process

A supercritical fluid is characterized as a dense-phase fluid with both pressure and temperature exceeding their respective critical values (see Fig. 1). Currently, supercritical CO_2 represents the predominant foaming agent in supercritical foaming technology, primarily due to its low critical temperature (31.1 °C) and moderate critical pressure (7.38 MPa) [45–48]. Supercritical fluids exhibit intermediate characteristics between gaseous and liquid states (see Table 1) [7, 49]. They possess diffusion coefficients similar to gases and solubility comparable to liquids [50–52].

Compared to supercritical gases such as N_2, supercritical CO_2 offers numerous advantages [53–56], including lower critical pressures and temperatures, as well as ease of preparation. It exerts a favorable plasticizing effect on polymer melts, capable of reducing the glass transition temperature and enhancing polymer solubility. Moreover, it can decrease the viscosity of polymer melts, improve fluidity, and lower processing temperatures. Additionally, CO_2 is environmentally friendly and non-toxic and leaves no residue from blowing agents, making it a widely utilized foaming agent in polymer foaming.

As illustrated in Fig. 2, the polymer foaming process involves the formation of the polymer/gas homogeneous system, cell nucleation, cell growth, and stabilization [57–61] (see Fig. 2). Under specific temperature and pressure conditions, gas diffuses into the polymer matrix, eventually reaching a state of dissolution equilibrium, creating a polymer/gas homogeneously saturated system. Subsequently, a rapid increase in temperature or a swift reduction in pressure leads the homogeneous system to attain a supersaturated state. This induces instability in the thermodynamic state within the system, serving as the driving force for cell nucleation [62–64]. Following this, the gas rapidly dissolves and diffuses into the cell nuclei, initiating continuous cell growth. This growth process involves continuous gas consumption and reduction. After a certain period, insufficient gas remains in the system to sustain further cell expansion, leading to a decrease in the driving force for cell growth. Meanwhile, as temperature decreases, the viscoelasticity of the polymer melt increases, leading to greater resistance to cell growth. Eventually, cell growth ceases, and the cell structure and density stabilize. A comprehensive exploration of the foaming mechanism can optimize polymer foaming processes, thereby enhancing the performance of polymer foam.

2.1.1 Formation of homogeneous systems

The creation of a polymer/gas homogeneous system is crucial in the microcellular foaming process. An inert gas diffuses and dissolves within the polymer under specific temperature and pressure conditions, leading to the formation of a uniform polymer/gas saturation system. The solubility of the gas is directly associated with the temperature and pressure of the system and can typically be estimated using Henry's law [65]:

$$C = K_h P \tag{1}$$

where C is the solubility of the gas in the polymer melt, P is the pressure, and K_h is the Henry coefficient. According to Henry's law, the solubility of a gas becomes constant at a specific pressure and temperature, indicating the gas's solubility limit in the polymer. To ensure the creation of a homogeneous system during foaming, the applied pressure is typically considerably higher than the saturation pressure of the gas. Thus, comprehending gas solubility is crucial [66–70].

In addition to gas solubility, the creation of a homogeneous system is influenced by the gas diffusion rate, which is primarily dictated by temperature. The diffusion coefficient of gases in polymers can be determined using the following equation [71]:

$$D = D_0 \exp\left(\frac{E_a}{R_g T}\right) \tag{2}$$

where E_a is the diffusion activation energy, R_g is the gas constant, and D_0 is a pre-exponential factor.

2.1.2 Cell nucleation

Cell nucleation is a critical stage in polymer foaming, influencing the cell density and cell size [72–74]. When a polymer/gas homogeneous system undergoes a rapid change in temperature or pressure, it causes a sudden decrease in solubility. This decrease results in a high degree of supersaturation, leading to the formation of numerous cell nuclei within the matrix. Controlling the process of cell nucleation significantly affects both the microstructure and properties of the foams.

Volmer and Weber introduced the classical nucleation theory in 1926. However, it is important to note that the application conditions of the traditional classical nucleation

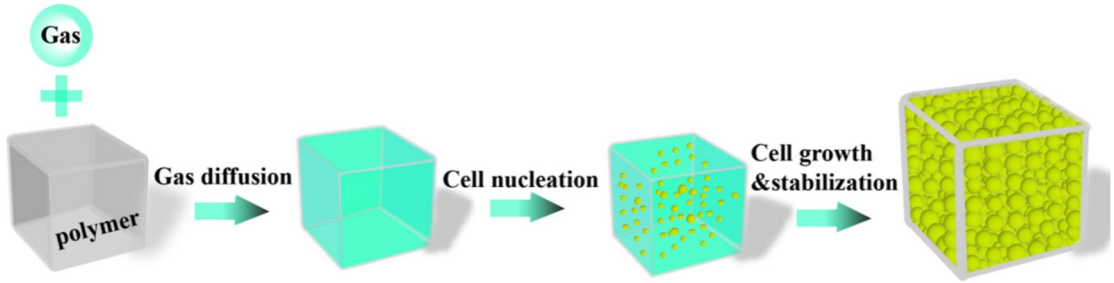

Fig. 2 Supercritical foaming process [47]

theory for general nucleation differ from those for microcellular foam. Therefore, Colton and Suh developed the cell nucleation theory for microcellular foam, expanding upon the classical nucleation theory [75, 76]. This theory categorized homogeneous and heterogeneous nucleation [77]. The free energies (G) can be calculated according to the following equation [78]:

$$G = \sum_i u_i N_i + \sum_j A_j \gamma_j - PV + SE \tag{3}$$

where μ_i is the chemical potential of the component, N_i is the number of molecules, γ_i is the interfacial energy of the interface, A_j is the interfacial area, P and V are the pressure and volume, respectively, and SE is the elastic energy of the system.

(1) Homogeneous nucleation.

Homogeneous nucleation occurs within a uniform system. In theory, every molecule can serve as a nucleation site; however, substantial free energy is needed for nucleation due to the lack of nucleation site attachment and induction, requiring a high degree of supersaturation. The Gibbs free energy (ΔG_{hom}) for homogeneous nucleation can be expressed with the following equation:

$$\Delta G_{hom} = -\frac{4}{3}\pi r^3 \Delta P + 4\pi r^2 \gamma \tag{4}$$

where γ is the surface tension, r is the cell radius, and ΔP is the pressure difference between the inside and outside of the cell. The first term on the right side of the equation denotes the volume free energy, while the second term represents the interfacial energy necessary for cell growth.

The maximum value of ΔG_{hom}, denoted as ΔG^*_{hom}, occurs at the critical cell radius R^*, where $\partial \Delta G_{hom}/\partial R = 0$. The critical radius for homogeneous nucleation is then expressed as the follows:

$$R^* = \frac{2\gamma}{\Delta P} \tag{5}$$

The activation free energy of the critical cell for homogeneous nucleation is expressed by Eq. 6:

$$\Delta G^*_{hom} = \frac{16\pi\gamma^3}{3\Delta P^2} \tag{6}$$

The homogeneous nucleation rate is expressed as follows:

$$N_{hom} = f_0 C_0 \exp\left(-\frac{\Delta G^*_{hom}}{kT}\right) \tag{7}$$

where f_0 is the frequency factor of gas molecules entering the critical cell nucleus, C_0 is the concentration of gas molecules, k is the Boltzmann coefficient, and T is the temperature.

From the above equations, it can be inferred that as the pressure difference between the inside and outside of the cell increases, both the critical cell radius and the critical free energy decrease. This implies that more gases can serve as nucleation sites within the polymer system, consequently increasing the number of nucleation events. According to the homogeneous nucleation rate equation, elevating the gas concentration or temperature can also boost the nucleation rate. Additionally, increasing the pressure difference will further augment the nucleation rate.

(2) Heterogeneous nucleation.

Heterogeneous nucleation refers to the preferential formation of cells at heterogeneous nucleation sites, such as impurity and additive surfaces [78]. The presence of a low-energy point at the junction of coexisting gas, liquid, and solid phases reduces the nucleation energy potential barrier, facilitating nucleation (see Fig. 3). The nucleation process is illustrated in Fig. 3.

For heterogeneous nucleation, the free energy of the system changes as follows:

$$\Delta G_{het} = -\left(P_D - P_C\right)V_b + \gamma A_{g1} + \gamma_{sg} A_{sg} - \gamma_{s1} A_{s1} \tag{8}$$

where ΔG_{het} is the change in free energy of the heterogeneous nucleation, P_D is the pressure inside the cell, P_C is the ambient pressure, and V_b is the volume of the cell nucleus. A_{gl}, A_{sg}, and A_{sl} are the areas of the cell-polymer, additive particle-cell, and additive particle-polymer interfaces, respectively. γ is the interfacial tension at the interface of the polymer and the cell, γ_{sg} is the interfacial tension between the solid and the gas phases, and γ_{sl} is the interfacial tension between the solid phase and the polymer melt.

The heterogeneous nucleation rate can be calculated as the follow equation:

$$H_{het} = f_1 C_1 \left(-\frac{\Delta G^*_{het}}{kT}\right) \tag{9}$$

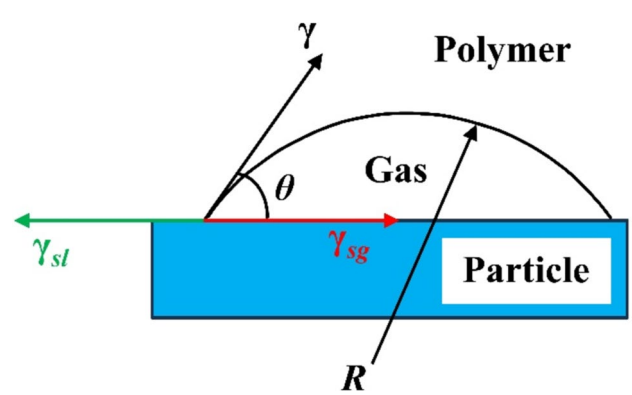

Fig. 3 Heterogeneous nucleation in the flat surface of solid particles

where f_1 is the frequency factor for heterogeneous nucleation, C_1 is the heterogeneous nucleating agent concentration, k is the Bozeman coefficient, and T is the temperature.

The $\Delta G^*{}_{\text{het}}$ can be calculated as the following equation:

$$\Delta G^*_{het} = \frac{16\pi}{3\Delta P^2}\gamma^3 S(\theta) \tag{10}$$

$$S(\theta) = \left(\frac{1}{4}\right)(2+\cos\theta)(1-\cos\theta)^2 \tag{11}$$

where ΔP is the saturation pressure of the gas and $S(\theta)$ denotes the geometric factor of the wetting angle. θ is the wetting angle at the interface of the polymer, additive, and gas. When θ equals π, the solid additive surface is entirely surrounded by the melt, rendering the additive inert. At this point, $S(\theta)$ equals 1, and ΔG_{hom} equals ΔG_{het}. Typically, in heterogeneous nucleation, the wetting angle falls between 0 and π, and the geometric factor $S(\theta)$ ranges from 0 to 1. For a typical wetting angle $\theta = 20°$, the order of magnitude of $S(\theta)$ is 10^{-3}, indicating that ΔG_{het} is significantly smaller than ΔG_{hom}. This suggests that heterogeneous nucleation is more favorable than homogeneous nucleation due to the lower energy barrier to overcome (see Fig. 4).

2.1.3 Cell growth

The growth of cells is a complex process involving heat transfer, mass, and momentum [79–83]. Due to the limitations of the experimental setups, most studies on growth kinetics rely on model simulations. The two primary cell growth models are the "Sea-island" model [84] and the cell model [85]. The "Sea-island" model describes the growth behavior of an individual cell in an infinitely large melt (see Fig. 5). However, it is only applicable under the ideal condition and does not accurately represent real cell growth scenarios. Therefore, while useful for theoretical purposes, the "Sea-island" model is not suitable for simulating real cell growth phenomena.

The sea-island model is predicated on several assumptions: (a) the bubble remains small during expansion, maintaining a spherical shape; (b) the width of the cavity is sufficiently large to allow unrestricted bubble expansion, unimpeded by cavity walls; (c) the polymer melt is considered incompressible; and (d) the viscosity of the polymer melt is exceedingly high, rendering buoyancy effects during bubble expansion negligible. By examining the growth dynamics of individual bubbles, this model elucidates the growth patterns of bubbles throughout the entire system. While it provides a useful approximation of the foaming process in polymer melts, the model has notable limitations in practical applications. First, assuming an infinite amount of gas dissolved in the melt around the bubble, and neglecting temperature variations, allows for continuous gas diffusion, theoretically leading to an infinite increase in bubble radius. Second, the model does not account for the interactions between bubbles during the expansion process, which diverges significantly from actual conditions. Consequently, the mathematical model derived from this framework requires further refinement and improvement to enhance its applicability to real-world scenarios.

The cell model elucidates the mass and momentum transfer within an individual cell and its encompassing shell layer of finite volume. It is assumed that the cell is enveloped by a concentric spherical shell of melt with a constant mass, known as the cell wall shell, containing a significant amount of dissolved gas (see Fig. 6). During the cell's expansion, gas diffusion solely occurs between the gas within the melt in the cell wall shell and the cell. This model addresses the constraints of earlier models by representing the foam body as comprising numerous spherical cells, offering a more realistic depiction of the melt in adjacent cells. It also provides

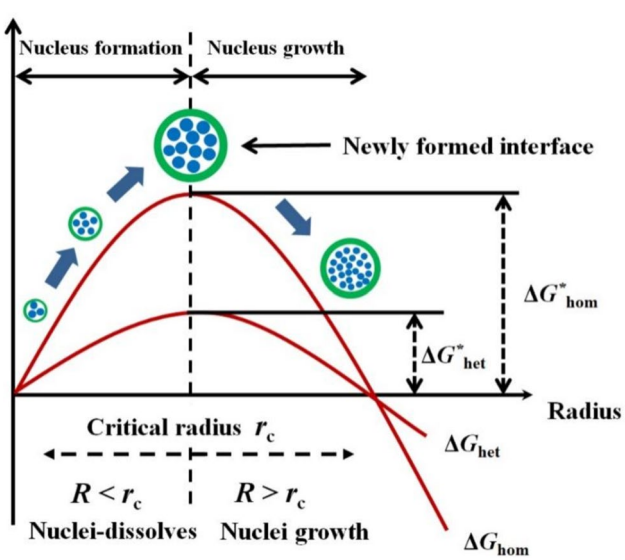

Fig. 4 Bubble nucleation and growth as a function of free energy

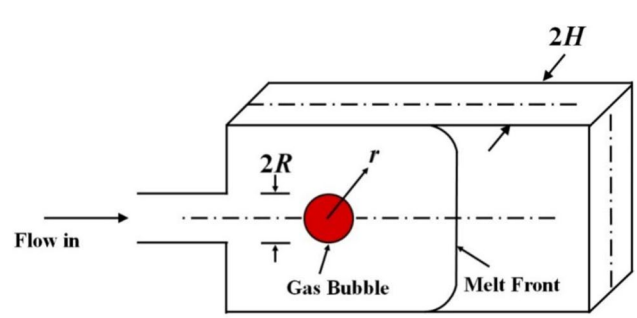

Fig. 5 "Sea-island" model

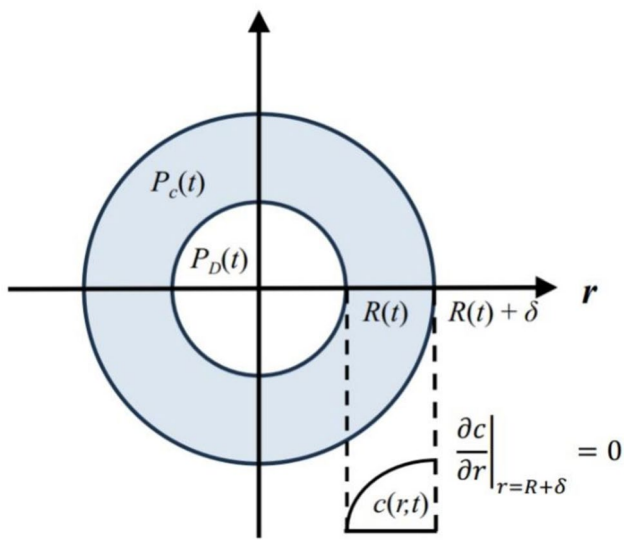

Fig. 6 Schematic of cell model

a more comprehensive solution to the interrelationships between the expansion processes of neighboring cells.

The most commonly employed physical model is the cell model, depicted in Fig. 6. As cells grow, gas from the surrounding melt diffuses continuously into the cell. This process results in a decrease in gas concentration within the melt, leading to the formation of a concentration boundary layer. Beyond this layer, the gas concentration stabilizes.

According to the cell model, the kinetics of cell growth under isothermal conditions can be explained by coupled mass and momentum conservation equations. The preservation of gas mass within the cell can be expressed as follows:

$$\frac{d}{dt}\left(\frac{4\pi}{3}\frac{P_D R^3}{R_g T}\right) = 4\pi R^2 D \frac{\partial c}{\partial r}\bigg| r = R \tag{12}$$

$$P_D - P_C - \frac{2\gamma}{R} + 2\int_R^{\infty} \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} dr = \frac{4\eta}{R}\left(\frac{dR}{dt}\right) \tag{13}$$

where R is the cell radius, $R = dR/dt$ is the cell growth rate, D is the diffusion coefficient, c is the gas concentration, η is the melt viscosity, R_g is the gas constant, and σ is the viscoelastic stress tensor. The viscoelasticity of the polymer is represented by the normal stress difference σ_{rr}-$\sigma_{\theta\theta}$. When comparing the two models, the cell model emerges as a more realistic representation of cell growth and is consequently extensively employed in simulating the cell growth process.

2.2 Effects of chemical structure and crystal structure

PP is a typical linear polymer known for its low melt strength. This characteristic becomes problematic during the foaming process, as the insufficient melt strength fails to withstand the stress of cell growth, leading to cell collapse and rupture [86–88]. Additionally, iPP has a narrow foaming temperature range, only 4 °C. As the melting temperature approaches the melting point of iPP, its melt strength sharply decreases, presenting a significant challenge to the foaming process. Moreover, being a semi-crystalline polymer, iPP encounters difficulties in achieving uniform cell morphology and structure compared to amorphous polymers, primarily due to the lower solubility of gases in the crystalline region [89–91]. During the iPP foaming process, cells tend to nucleate preferentially at the interface between the crystalline and amorphous regions, resulting in high nucleation density. This preference arises from the interface providing sites for heterogeneous nucleation with relatively low nucleation energy barriers.

2.3 Different methods to improve PP foaming properties

To improve the foaming quality of PP, current research primarily focuses on enhancing PP foaming performance through material modification [92–96]. This modification primarily involves blending, branching, cross-linking, and the incorporation of fillers.

2.3.1 Effect of blend

Blending modification [97–101] involves mixing different polymers to create blends, enhancing mechanical and processing properties, while potentially reducing material costs. The microstructure of blends, including crystal structure and phase distribution, can be modified to influence material properties, with such adjustments exerting a significant impact on overall performance. Polymer blends present a pivotal pathway for achieving high performance, refinement, functionalization, and the development of novel polymer material varieties [102–106].

Qiang et al. [107] investigated the foaming properties of PDMS/PP composites by blending polydimethylsiloxane (PDMS) with PP, followed by $ScCO_2$ foaming at various temperatures (see Fig. 7). The investigation revealed that PDMS enhanced PP's chain mobility and facilitated crystalline nucleation, leading to a significant increase in cell density in the blends. This effect was observed across a wider temperature range, ensuring the maintenance of a favorable cell structure. The research showcased the successful enhancement of PP's foaming performance through blending with PDMS.

You et al. [108] synthesized PP/HMSPP foam composites via blending modification and evaluated their mechanical properties (see Fig. 8). The investigation unveiled that increasing HMSPP content enhanced the crystalline

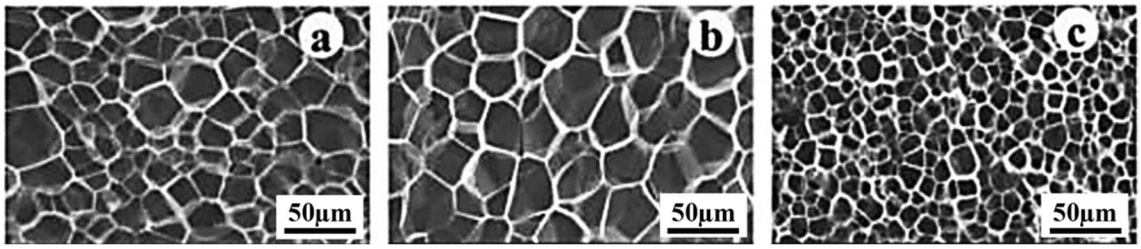

Fig. 7 Cell morphology of PP/PDMS composites at 145 ℃ and 15 MPa. **a** PP, **b** PP/PDMS770, and **c** PP/PDMS4000 [107]

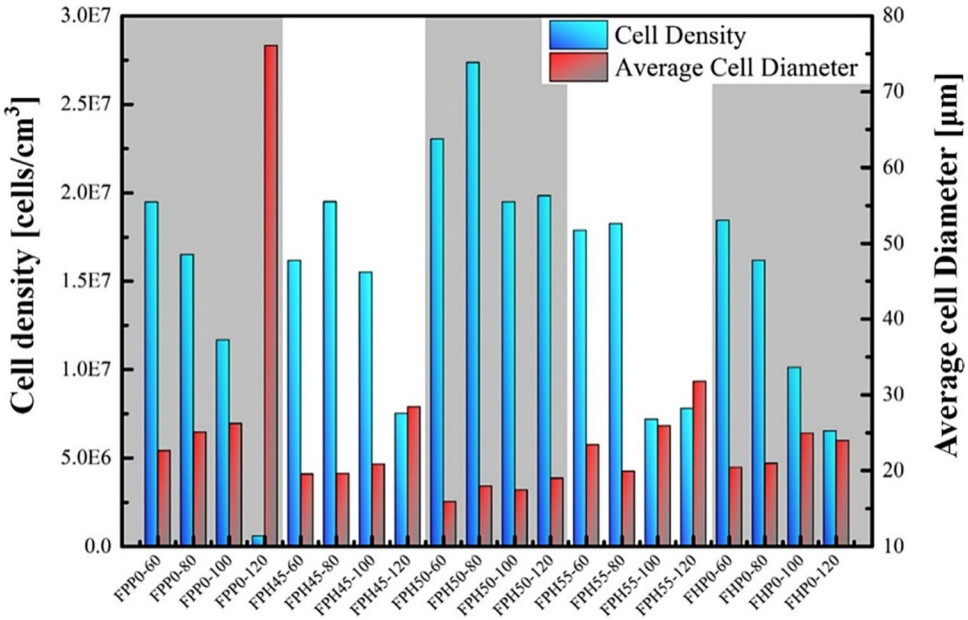

Fig. 8 Variations of average cell diameter and cell density of PP0, HP0, and PP/HMSPP composites foam at different mold temperatures [108] (FPP0-60 represents FPP0 prepared at a mold temperature of 60 ℃)

behavior of the composites, elevating both crystalline onset and peak temperatures. Moreover, a significant rise in melt viscosity was noted, while rheological behavior remained consistent across different mold temperatures. Consequently, blending with HMSPP markedly improved the cell quality of PP, leading to enhanced surface morphology and mechanical properties, particularly at 50 wt.% HMSPP content.

Yeh et al. [109] developed oil-absorbent open-cell foams by blending high-melt-strength polypropylene (HMSPP) and linear polypropylene (LCB-PP) in varying ratios using $ScCO_2$ at different temperatures and pressures. The findings indicated that higher HMSPP content resulted in an increased foam expansion rate and open-cell content, thereby broadening the foaming temperature range (see Fig. 9). Oil absorption experiments revealed a direct correlation between foam expansion ratio and oil absorption, which was inversely proportional to oil viscosity. Furthermore, cyclic adsorption experiments validated the reusability of the composites. Blending LCB-PP with HMSPP increased the strain hardening coefficient, thus expanding the foam's processing range and increasing its expansion ratio. Additionally, both the expansion rate and open-cell content showed an increase with the rise in saturation pressure.

2.3.2 Effect of branch

Branching modification [110–113] is a method of enhancing polymer properties by introducing a branched structure. Typically, in this process, the polymer is usually given a branched structure by introducing branched chains or branched groups into the polymer molecular chain. The introduction of branched chains alters the spatial structure of the polymer molecules and intermolecular interactions, effectively hindering the movement of the molecular chains and reducing their alignment density. As a result, the mechanical and processing properties of the materials are improved. [36, 114–116].

Mendoza-Cedeno et al. [117] investigated the effect of foaming temperature and molecular structure on the cell morphology and mechanical properties of PP/LCB-PP under varying activation temperatures and filling times of chemical blowing agents. Results showed that at the highest foaming

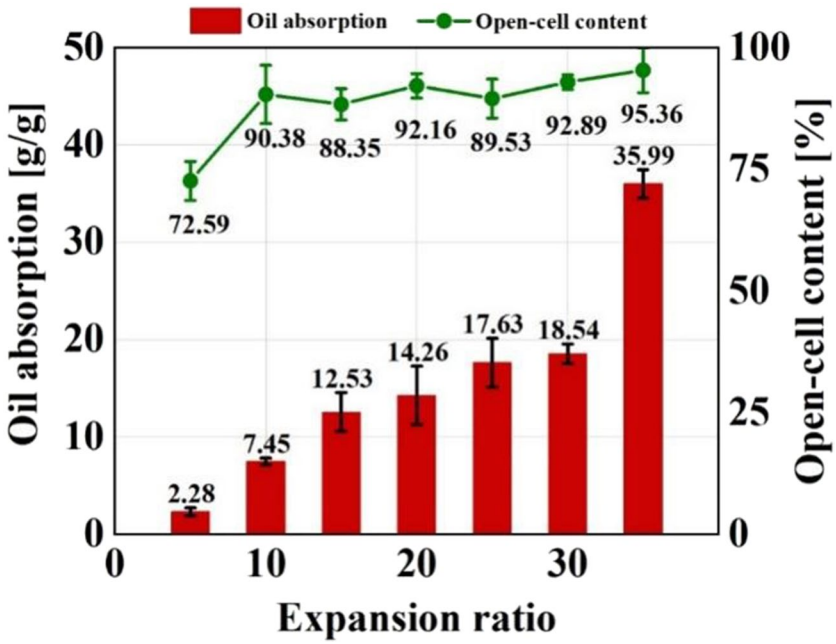

Fig. 9 Open-cell content and oil absorption of HMSPP/PP foam with different expansion rates [109]

temperature, PP/LCB-PP foams exhibited the 887% higher cell density, 46 wt.% smaller cell size, and more uniform cell structure compared to linear PP (see Fig. 10). Additionally, the flexural modulus of PP/LCB-PP foams increased by 44 wt.% compared to linear PP. These findings suggest that the branched structure of PP/LCB-PP foams contributes to enhancing cell morphology and mechanical properties under high-temperature injection molding conditions.

2.3.3 Effect of cross-linking

Cross-linking modification [118–121] involves linking linear or branched polymer chains through covalent bonds to create a three-dimensional network, commonly known as chemical cross-linking. This process introduces extra attachment points, enabling polymer chains to connect via covalent bonds or physical cross-linking structures. The resulting interconnected molecular chains form a three-dimensional network that enhances material strength, hardness, and thermal stability, while also influencing properties like shape memory and elasticity [122–125].

Yang et al. [126] blended various proportions of PP and high-density polyethylene (HDPE), induced cross-linking by γ-ray irradiation, and prepared composite foams using $ScCO_2$ foaming. The study investigated the impact of irradiation cross-linking on the melting point and crystallinity of the composites. It was observed that the presence of the HDPE phase facilitated the dissolution of CO_2 in the blends. Furthermore, the tensile strength and elongation at break of PP/HDPE cross-linked foams surpassed those of pure PP. Additionally, irradiation cross-linking of the HDPE phase enhanced the melt strength of the PP/HDPE foams,

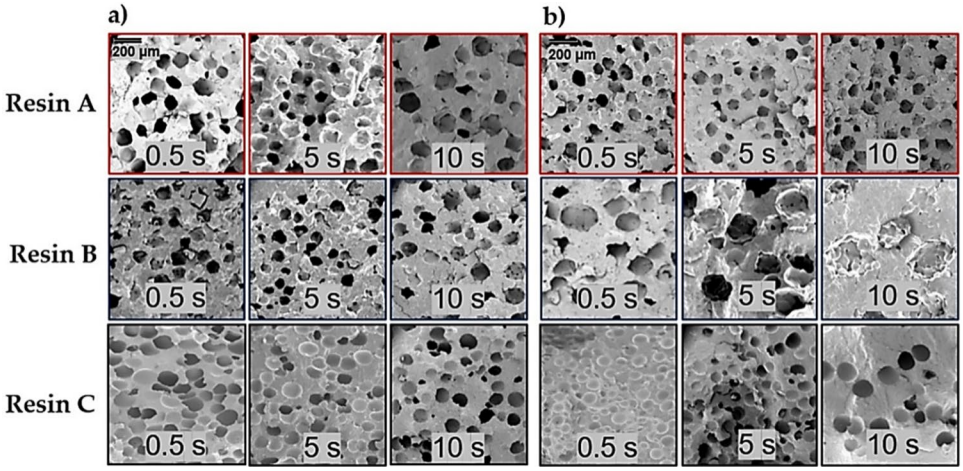

Fig. 10 Cell morphology of A (linear PP), B (BMWD-PP), and C (BMWD-branched PP) when foamed with **a** 2 wt% AV CBA (a low-activation-temperature chemical blowing agents) and **b** 2 wt% UC CBA (a high-activation-temperature chemical blowing agents) at different filling times (0.5, 5, and 10 s) [117]

consequently widening the foaming temperature range for PP (see Fig. 11).

In addition, Yang et al. [127] prepared composite foams by blending different proportions of PP with styrene-ethylene-butadiene-styrene (SEBS) using γ-ray induced cross-linking and employing $ScCO_2$ foaming. The study revealed that the melt strength of PP/SEBS foams obtained through radiation cross-linking was significantly enhanced. These foams exhibited smaller cell size, higher cell density, greater closed cell rates, and increased tensile strengths (see Fig. 12). Furthermore, the foaming temperature range of the cross-linked PP/SEBS foams expanded from 4 to 10 °C compared to that of pure PP foams. Thus, the incorporation of SEBS and radiation cross-linking markedly improved the foaming properties of PP.

2.3.4 Effect of filler

Mehdipour et al. [128] investigated the effect of nanoparticles on the rheological and mechanical properties of LPP (linear polypropylene) using two different aspect ratios of nano-$CaCO_3$ and nanoclay as fillers. They observed that nano-$CaCO_3$ enhanced the tensile strength, elongation at break, and tensile modulus of the composites. Additionally, during the foaming process, nanoclay increased foam density, reduced cell size, and enhanced melt strength. The combined presence of nanoclay and calcium carbonate nanoparticles further elevated foam density. This study illustrates that incorporating nanoclay fillers into LPP significantly improves its foaming properties.

Mark et al. [129] fabricated fiber-reinforced composites by incorporating polyethylene terephthalate (PET) nanofiber fillers into polypropylene. The study revealed that PET nanofibers not only enhanced the tensile and flexural properties of solid (unfoamed) injection molded parts, but also substantially ameliorated the cell morphology and cell structure of the PP/PET composite foam. This improvement led to a more than three orders of magnitude increase in cell density and enhanced cell uniformity (see Fig. 13). Furthermore, the tensile and flexural properties exhibited enhancement with increasing PET content, resulting in a doubling of foam strength and flexural strength by up to 100%.

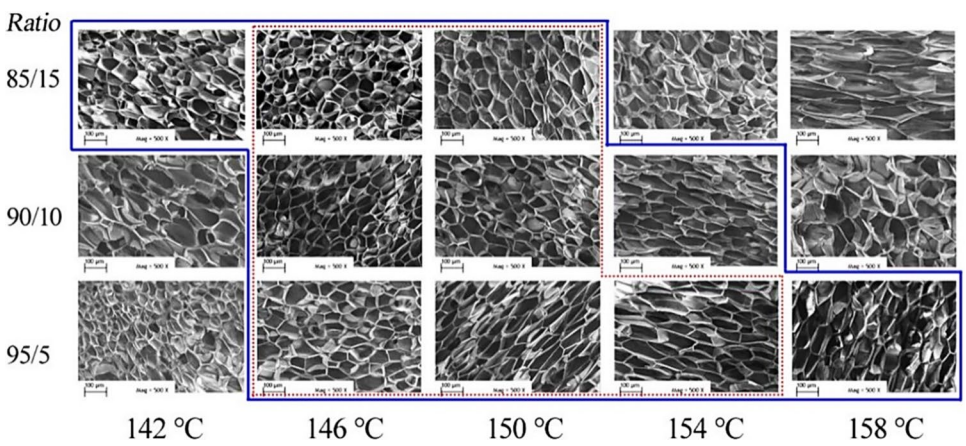

Fig. 11 SEM of PP/HDPE composite foam under different foaming conditions at 20 MPa [126]

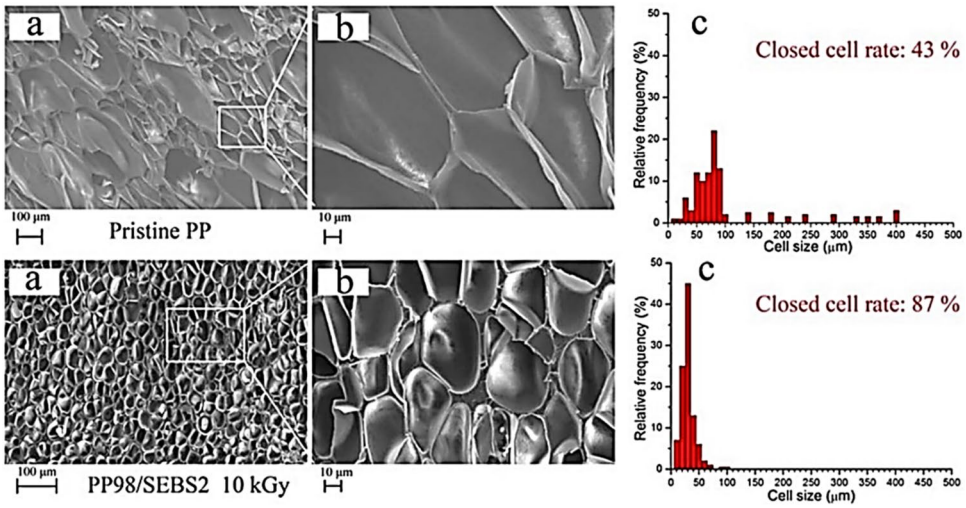

Fig. 12 SEM of PP foam and irradiated PP98/SEBS2 foam: **a** low power image and **b** high power image. **c** Cell size distribution [127]

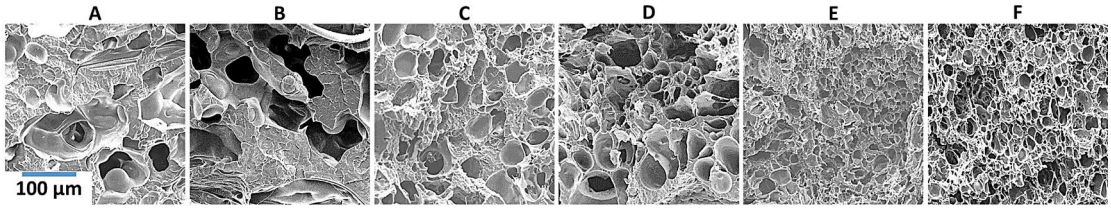

Fig. 13 The foam SEM micrographs with **A** as-received PP, **B** 0 wt% PET, **C** 1 wt% PET, **D** 5 wt% PET, **E** 10 wt% PET, and **F** 15 wt% PET [129]

Liu et al. [130] synthesized PP/GN nanocomposites by incorporating graphite nanosheets (GN) into PP and explored the influence of GN on the crystallization and rheological behavior of the composite. The investigation unveiled that GN incorporation escalated the nucleation rate and crystallinity, substantially shrinking the average cell size by a factor of at least seven and elevating the cell density by two orders of magnitude. Moreover, the inclusion of GN enhanced the melt viscosity of PP. The specimens, crafted using the mold-opening foam injection molding (MOFIM) process, exhibited a microporous structure with an average diameter of a few micrometers, thereby augmenting the mechanical properties of the materials. The tensile strength and modulus of the PP/GN nanocomposite foams increased by nearly 50% and 70%, respectively, compared to pure PP, while the impact strength surged by over 75%.

3 PP foam manufacturing techniques

The foaming techniques currently employed for PP include batch foaming, extrusion foaming, and injection molding. Batch foaming is ideal for generating limited sample quantities, extrusion is utilized for continuous profile production, and injection molding is preferred for materials with intricate shapes. The advantages and drawbacks of these techniques depend on the specific application requirements. For instance, batch foaming is suitable for small-scale research material production, while injection molding is better suited for mass production involving complex processes. Extrusion foaming enables continuous production but necessitates significant initial capital investment in equipment.

3.1 Batch foaming

Batch foaming is the predominant method for preparing microporous plastics [131–134]. In this process, under specific temperature and pressure conditions, gas diffuses into the polymer in a high-pressure reactor until saturation is reached. Rapid pressure reduction or temperature increase induces thermodynamic supersaturation, leading to cell nucleation. Gas dissolves and diffuses into the cell nucleus, promoting cell growth, followed by rapid cooling to form the stable cell structure. Batch foaming offers advantages such as a high nucleation rate, uniform cell morphology, and high cell density. However, batch foaming lacks continuous production capability, leading to low production efficiency. Consequently, batch foaming is primarily utilized to produce limited sample quantities.

Currently, research on the batch foaming process primarily focuses on understanding the influence of supercritical fluids, process parameters, and foaming routes on the foaming behavior of polymers. Guo et al. [140] explored the foaming behavior of PP using commonly employed supercritical fluid blowing agents, namely, CO_2 and N_2, under varying experimental conditions. Their findings indicated that, under identical experimental conditions, PP foam produced with CO_2 as the blowing agent had a higher cell density compared to that produced with N_2. Hou et al. [135] employed a cooled batch foaming method to produce high-expansion PP foams, investigating the interaction between crystallization and foaming behavior of PP. They observed significant changes in crystal nucleation and onset crystallization temperature of PP under the influence of CO_2. Within the pressure range of 0 to 10 MPa, increasing CO_2 pressure led to larger cell sizes. Further increasing the pressure to 15 MPa resulted in higher cell density. Moreover, under CO_2 conditions, the crystallization rate of PP was lower compared to conditions without CO_2, indicating that dissolved CO_2 impedes crystal nucleation. The onset crystallization temperature exhibited a linear decrease with increasing CO_2 pressure within the pressure range of 0 to 15 MPa. Kumar et al. [136] investigated PP foams produced through two different foaming methods: single-step pressure-induced foaming and two-step temperature-induced foaming, both conducted under low saturation pressure. The study systematically examined the effects of these two batch foaming methods on cell morphology and mechanical properties. The decrease in the stiffness and compressive strength of the resulting foams with the augmentation of cell size was observed in both methods. Additionally, the crystallinity of the foams decreased with higher foaming temperature. Overall, the thermal stability of foams produced via both

pressure-induced and temperature-induced foaming methods showed improvement compared to non-foamed PP.

3.2 Extrusion foaming

Extrusion foaming [137–140] enables the rapid preparation of PP microcellular foams by integrating a gas injection system, a heating system, and a specialized foaming die head. In this continuous process, polymer raw material is fed from a hopper into a screw extruder, where it undergoes melting and plasticization due to the shearing action of the screw and heating from the heater. The blowing agent is then introduced into the melt within the extruder at a specified flow rate. Subsequently, the foaming agent is uniformly dispersed into the polymer matrix through the mixing device of the die, forming a homogeneous polymer/gas system. Rapid depressurization occurs under the action of the nucleation nozzle, inducing significant thermodynamic instability and resulting in the formation of a large number of cell nuclei. Finally, the foam undergoes cooling in the molding unit to complete the process. Compared to batch foaming, extrusion foaming offers superior economic benefits such as high output, easy regulation, and versatility in producing various product shapes. However, it demands advanced equipment and entails higher initial capital investment for equipment acquisition.

Huang et al. [141] proposed an innovative method for continuously producing expanded PP foam beads, employing supercritical CO_2 extrusion foaming. This approach streamlines material capacitation, compounding, granulation, and foaming into a single-step process, notably reducing production time and energy consumption. The resulting foamed products exhibited exceptional mechanical and thermal insulation properties. Su et al. [142] fabricated composite foam beads comprising PP and thermoplastic polyurethane (TPU) using a comparable approach of supercritical CO_2 extrusion foaming combined with air-cooled pelletizing technology. By adjusting the rotational speed of the tool, they could vary the foam bead geometries, leading to an increase in cell size with higher tool speeds. Additionally, integrating TPU significantly improved the cell structure of the composite foam beads, resulting in more uniform cell size. Tammaro et al. [143] explored the foaming properties on atactic polypropylene (aPP) through foam extrusion. The optimal foaming temperature was determined by varying the concentration of CO_2. The study revealed that the density of PP foams decreased as the concentration of the blowing agent increased. PP foam beads exhibited sintering within a wide range of vapor pressures (0.25 ~ 0.3 MPa) and cycling times (110 ~ 228 s). Sintering of PP could be achieved at a pressure of 3 bar, leading to reduced energy consumption during manufacturing. Hoe et al. [144] employed lignin as a nucleating agent in the supercritical CO_2 extrusion foaming of HMSPP and examined the foaming properties of the composite. Compared to PP foams, PP/lignin foams exhibited superior energy absorption capacity at the same density, offering a promising method for producing environmentally friendly HMSPP foams.

3.3 Injection foaming

In the injection foaming [145–149] process, the polymer and blowing agent are blended in the screw barrel to create a homogeneous system. This mixture is then injected through the nozzle into the mold, where a thermodynamic imbalance rapidly occurs due to the low temperature and pressure of the mold, initiating the nucleation of cell that grow and set in the mold. Compared to the extrusion foaming process, injection foaming offers unique advantages. For example, the ability to create forming molds with specific required shapes enables the production of parts with complex geometries.

Zhao et al. [150] employed injection molding to fabricate in situ fibrillated PP/PTFE composites, resulting in a substantial enhancement in the crystallinity and viscoelasticity of the PTFE fibers. This improvement led to a significant refinement of the cell structure in the PP/PTFE composite foam, accompanied by notable improvements in mechanical properties and increased impact strength with higher PTFE content. Additionally, the incorporation of PTFE fibers enhanced the tensile strength and modulus of the PP foams. Furthermore, Zhao et al. [151] utilized a combination of microcellular injection molding (MIM) and in situ fiber trembling techniques to create lightweight, lipophilic, and acoustically insulating PP/PTFE composite foams. The PTFE nanofibers obtained through the in situ fibrillation method significantly improved the crystallinity and melt strength of the PP matrix. This improvement effectively enhanced the foaming ability of PP, resulting in an increased expansion ratio to 20.2 and a density of 10^8 cells/cm^3. The substantially refined cell structure and larger expansion ratio markedly boosted the oil absorption capacity of the PP/PTFE composite foam to 22.5 g/g, with a recovery rate as high as 97.4%. Moreover, the PP/PTFE composite foam exhibited robust absorption capability for various types of oil. Concurrently, the sound absorption coefficient and transmission loss of the PP/PTFE composite foam were significantly improved across different sound frequencies, with sound absorption coefficients surpassing 0.5 and transmission loss reaching up to 50 dB.

4 Application

Polypropylene is renowned as a versatile plastic, celebrated for its affordability, high heat distortion temperature, and outstanding chemical resistance. It has garnered significant

attention in foam materials research in recent years. PP finds extensive applications in automotive manufacturing, flame retardant systems, conductive and electronic shielding, oil and water separation, food packaging, and various other fields. Nonetheless, challenges arise in the foaming process due to its low melt strength, which hinders the attainment of the desired cell structure, limiting its applicability in foaming applications.

4.1 Automotive

Recently, the automotive industry has increasingly focused on utilizing green technologies to manufacture automotive parts, reduce production costs, and enhance fuel efficiency [152]. Employing polymer foams has emerged as an effective strategy to address this challenge [153, 154]. Among various polymer foams, there is a growing demand for PP in the automotive sector owing to its economic viability, weight reduction benefits, excellent mechanical properties, ease of processing, and recyclability. Thus, PP foams find extensive use in manufacturing components like car bumpers, advanced safety car seats, door shock cores, car side shock cores, dashboards, cushioning pads, headrests, and toolboxes [155, 156].

Ding et al. [157] fabricated modified electrically insulating polypropylene (EiPP) composite foams, termed mEiPP, by coating EiPP with CNTs@sPP powders. This involved using low-melting-point syndiotactic polypropylene (sPP) powder coated with carbon nanotubes (CNTs) as a microwave absorbing medium. The process integrated supercritical carbon dioxide foaming and microwave sintering techniques to establish a conductive three-dimensional skeleton. The samples exhibited excellent mechanical properties, boasting a tensile strength of 4.07 MPa and a compressive strength of 4.64 MPa. Moreover, at a CNTs content of only 0.26%, the conductivity and short-circuit current measured 2.83×10^{-4} S/m and 3.0 nA, respectively. These values met the criteria for antistatic properties, ensuring safety for automotive interior applications. Ozunlu et al. [158] employed graphene nanosheets (xGnP) as additives in the foam injection molding process to modulate the foaming of copolymer polypropylene (co-PP). They investigated the effects of two different concentrations of xGnP on the properties of PP/xGnP composites. The study found that incorporating xGnP improved the flowability, mechanical properties, and foaming characteristics of co-PP. Compared with the samples prepared using the conventional method (twin-screw extrusion followed by foam injection molding), the nanocomposites containing 2% xGnP showed a twofold reduction in cell size and exhibited significant enhancements in flexural modulus and flexural strength, increasing by 47% and 122%, respectively. Moreover, there was a notable 40 dB increase in acoustic transmission loss performance compared to pure foam, indicating improved low-frequency acoustic transmission loss. Additionally, the samples showcased a 12% weight reduction, meeting the weight reduction requirements and mechanical property standards for automotive parts .

4.2 Flame retardant

Polypropylene, being a linear thermoplastic polymer, is flammable and emits a substantial amount of heat and toxic smoke during combustion, which poses a risk of serious fire accidents and restricts its widespread application. Therefore, the development of an ultra-lightweight, ultra-strong polypropylene foam with high flame-retardant properties is of paramount importance [159–164].

Shi et al. [165] synthesized PP/EG/CNTs/IFR composite foams by incorporating an intumescent flame retardant (IFR) along with expandable graphite (EG) and CNTs, investigating their flame retardant properties. The study revealed that the combination of IFR, expanded EG, and CNTs synergistically improved the flame retardancy of PP/EG/CNTs/IFR composite foam. Compared to PP/IFR composite foam, the PP/10EG/5CNTs/25IFR composite foam exhibited enhanced flame retardancy, superior ablative resistance, and excellent thermal insulation across various flame conditions. Additionally, during combustion, the PP/EG/CNTs/IFR composite foam material developed a complete porous carbon layer on its surface, effectively preventing droplet phenomena and preserving the overall shape of the composite foam. Huang et al. [166] utilized supercritical carbon dioxide foaming followed by defoaming to fabricate PP/IFR composites, aiming to enhance dispersion homogeneity. The flame retardant and mechanical properties of these composites were subsequently examined. The investigation demonstrated that the combination of supercritical carbon dioxide plasticization and biaxial cell stretching led to an increased foam expansion ratio and enhanced cell stretching force, thereby significantly improving the dispersion of IFR particles within the composite. As a result, the flame retardancy and mechanical properties of the PP/IFR composites were notably enhanced due to the improved dispersion of the IFR particles. Jiang et al. [167] employed a combination of in situ fibrillation (INF) and supercritical carbon dioxide foaming technology to fabricate composite foams consisting of in situ fibrillated PP/PET/poly(diarylidene phosphoronitrile) (PDPP) and subsequently investigated their mechanical and flame retardant properties. The findings indicated that the PP/PET(F)/PDPP foam exhibited a finer cellular structure compared to the pure PP foam, with the cell size decreasing from 69 to 23 μm and the cell density increasing from 5.4×10^6 to 1.8×10^8 cells/cm^3. Furthermore, the PP/PET(F)/PDPP foam displayed notable mechanical properties, with the compressive stress increasing by 975%. The synergistic effect of the PET nanofiber network and the low loading of the PDPP

additive inhibited the combustion process, thereby enhancing the flame retardancy of the composite foam (see Fig. 14).

4.3 Conductive and electromagnetic shielding

With the widespread use of modern electronic products, electromagnetic radiation has emerged as a new environmental hazard, highlighting the growing significance of researching and developing materials with excellent electromagnetic shielding properties [168, 169]. Conductive polymer composites (CPCs) are preferred in the field of electromagnetic shielding due to their lightweight nature, corrosion resistance, ease of molding, and the flexibility to regulate performance [170–176].

Li et al. [177] produced large-size PP/carbon black (CB) composite foams measuring $60 \times 30 \times 10$ cm^3 using supercritical carbon dioxide blowing combined with steam-chest molding. The investigation unveiled that the prepared PP/CB foam featured a closed-cell structure with cell sizes in the hundreds of micrometers and a density ranging from 0.034 to 0.117 g/cm^3. At a CB content of 30%, the foam exhibited high conductivity of 1.3 S/m and electromagnetic interference shielding efficacy (SE) of 20–31 dB across a broadband frequency range of 6–18 GHz, shielding 99 to 99.92% of electromagnetic energy. Increasing the thickness of the samples to 4 to 6 cm resulted in a high SE of 41 dB (99.992% shielding power) in the range of 8–12 GHz. Additionally, all PP/CB foams exhibited an absorption-based shielding mechanism, with absorption coefficients reaching as high as 0.80–0.97 at 25–30% CB content, absorbing 80–97% of the power. Fu et al. [171] produced an iPP/CNT nanocomposite foam with a sandwich structure using a fusible blending process and the supercritical carbon dioxide foaming method. This method aimed to leverage the varying viscoelastic properties of blends of iPP and CNTs with different CNT contents at a specified foaming temperature. The investigation revealed that the bulk conductivity of the nanocomposites increased gradually as the iPP particle size decreased. For instance, the bulk conductivity of iPP-3 CNT composites prepared using iPP particles with a diameter of 3.5 mm was approximately 8.6×10^{-6} S/cm, whereas it increased to about 1.4×10^{-3} S/cm when the iPP size was 0.8 mm, representing more than a 150-fold increase compared to the 3.5-mm case. Moreover, the fabricated sandwich structure comprises a microfoam core layer (with low carbon nanotube content) and a nanofoam skin layer (with high carbon nanotube content), potentially enhancing the synergistic effect on the electromagnetic interference (EMI) shielding performance. The total electromagnetic shielding efficiency of the foam, denoted as iPP-10CNTs/iPP-3CNTs/iPP-10CNTs, is 38.0 dB, indicating superior EMI shielding performance compared to single-layer microfoam or its solid counterparts. Wu et al. [178] employed the back-core foaming injection molding technique to prepare polypropylene/

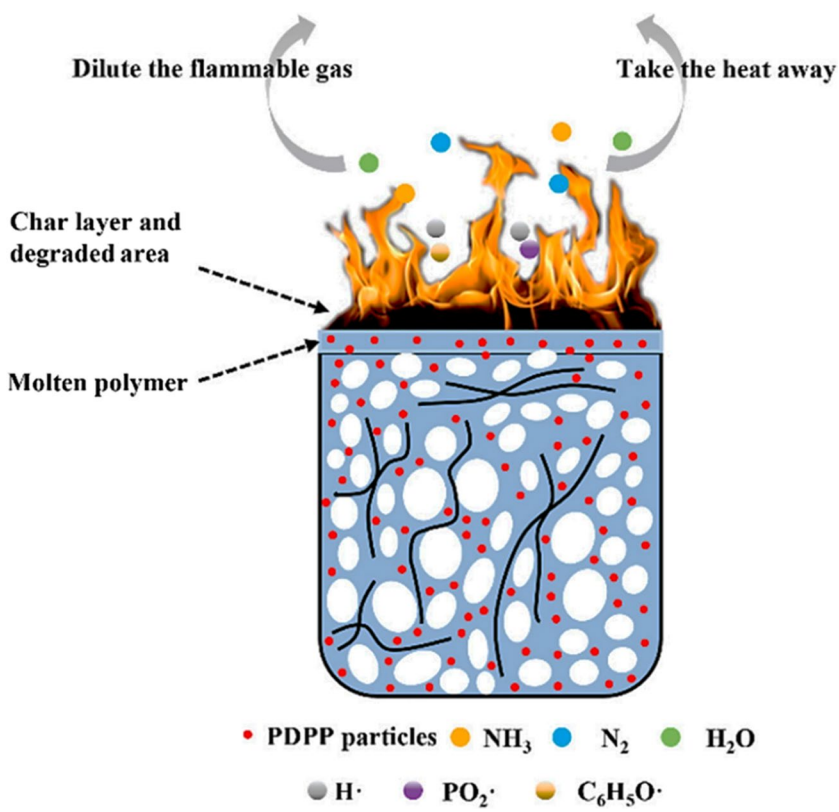

Fig. 14 Flame retardant mechanism of PP/PET(F)/PDPP foam [167]

carbon nanostructured (PP/CNS) nanocomposite foams with consistent porosity. They adjusted the nitrogen content to control the cell size, thereby notably improving the electrical properties of the conductive polymer composite (CPC). The research unveiled that as the cell size increased, the electrical conductivity of the PP/CNS nanocomposite foam escalated, achieving an electromagnetic interference efficiency of 329 dB cm2/g. This outcome showcased superior electromagnetic shielding performance compared to other CPC foams.

4.4 Oil–water separation

In recent years, the indiscriminate discharge of waste oil, industrial dirty oil, and oil spills at sea has not only significantly damaged the natural ecological environment but has also posed a threat to human health [179–182]. Consequently, the rapid and effective management of oil pollution has emerged as a pressing concern for researchers worldwide. This urgency is particularly evident in the advancement of oil–water separation materials featuring an open-cell structure [183, 184].

Zhao et al. [185] prepared a novel microporous PP/CNTs/sorbitol nanocomposite foam for oil-absorption and oil/water separation using injection foaming technology. The PP nanocomposite foam exhibited a cell size of only 20 μm, with a cell density reaching up to 10^{10} cells/cm^3, and an expansion ratio of 55.2 (see Fig. 15). Additionally, the unique layered porous structure of the PP nanocomposite foam resulted in an exceptionally high open-cell content of up to 99.4%, a water contact angle of 151.3°, and an adsorption capacity of up to 41.2 g/g for crude oil. Huang et al. [186] employed a multiphase polypropylene/CNTs/sorbitol nanocomposite foam for oil absorption and oil/water separation. The layered porous structure was found to significantly enhance oil removal capacity while maintaining high oil–water separation efficiency. The presence of additional secondary smaller voids significantly enhanced oil removal capacity compared to single voids. Moreover, the complete interconnection among secondary cell, primary cell, and additional oil flow channels resulted in superior oil–water separation performance. Su et al. [187] developed PP/poly(lactic acid) (PLA) composite foams with controllable hollow radial gradient porous structures using supercritical carbon dioxide-assisted foam-coupled extrusion and stretch-assisted techniques. By adjusting the extrusion foaming temperature and pressure, the fabricated foams exhibited a structure comprising dense skin layers, hollow cores, and closed and open cell. The outer diameter could be varied from approximately 9.8 to 5.7 mm, while the inner diameter ranged from about 5.5 to 1.7 mm. The investigation demonstrated that the prepared PP/PLA foams had a water contact angle of 146.2°. Moreover, when passing the oil–water mixture through the pressure filtration device, water swiftly passed through the hollow composite foams, while oil was absorbed. This observation underscores the excellent oil–water separation performance of the PP/PLA composite foam, indicating promising prospects for its application in wastewater treatment.

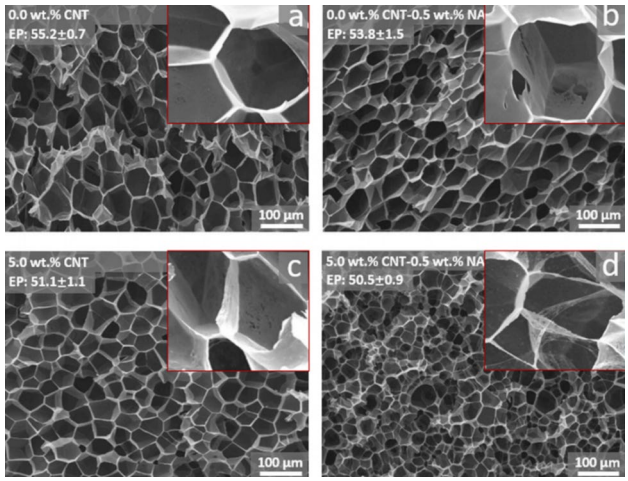

Fig. 15 SEM of PP/CNT/ sorbitol derivative nanocomposite foam [185]

4.5 Food packaging

Polypropylene has become indispensable in the food packaging sector due to its remarkable physical and chemical properties and cost-effectiveness. However, the pressing need for environmentally friendly PP composites has arisen in light of increasing environmental pollution and energy crises [188–191].

Tsagdi et al. [192] proposed a novel method for preparing food-grade polypropylene-based foams coated with blowing agents. This approach involved coating PP pellets with sodium bicarbonate and organic acids, with the addition of poly(ethylene oxide) (PEO) or aliphatic esters as coating stabilizers. Various combinations of PP and blowing agents were evaluated using a laboratory-scale injection molding machine, followed by testing selected combinations on a large industrial extruder. The investigation revealed a significant 14% increase in porosity and a 20% decrease in thermal conductivity of the polypropylene products. These findings contribute to cost reduction and improved thermal insulation of food-grade polypropylene foamed products.

5 Conclusion and perspectives

This paper offers a comprehensive review of various methods used to enhance the properties of polypropylene foam, polypropylene foaming technologies, and their applications.

The study illustrates that modifying polypropylene can improve its foaming properties and melt strength by altering its crystal and molecular structure. The melt strength was increased by adjusting the crystal structure and phase distribution of blends, thus improving the foaming and mechanical properties of the materials. Branched modification significantly improves foam structure and mechanical properties by adjusting the polymer molecular structure. Cross-linking modification enhances melt strength and thermal stability by creating a three-dimensional mesh structure between molecular chains. Additionally, the addition of fillers facilitates cell nucleation, thereby improving the foaming properties and thermal stability of polypropylene.

Samples produced via batch foaming demonstrate high cell density, small cell size, and superior mechanical properties. However, the extended time needed for gas saturation impedes continuous production, rendering it inadequate for industrial mass production requirements. The extrusion foaming efficiently generates PP foams with desirable cell structure and outstanding mechanical properties by integrating a gas injection system, a heating system, and a specialized foaming die. Nonetheless, the equipment entails high costs and necessitates a significant initial investment. Injection foaming employs a customizable, complex mold, making it suitable for fabricating foamed products with unique shapes.

PP foam material holds significant competitive advantages across various applications, particularly in automotive and flame-retardant fields, offering potential as a substitute for existing PS and PE foam materials. In automotive manufacturing, the utilization of lightweight and cost-effective PP composite foams for automotive interior and components aligns with the industry's demand for lightweight components and cost control. In the realm of flame retardancy, PP possesses a natural advantage attributed to its low thermal conductivity and excellent thermal insulation properties. Through compounding with fillers such expandable graphite and carbon nanotubes, cost savings can be realized, and simultaneously, excellent thermal stability and flame retardancy can be achieved. Moreover, in the domain of electromagnetic shielding, PP blended with CB or CNTs exhibits outstanding shielding properties against electromagnetic interference. In the realm of oil–water separation, PP composite foams demonstrate high oil absorption capacity, excellent oil–water selectivity, and the capability for repeated use, positioning PP as a promising solution for addressing oil–water separation challenges.

Despite the considerable promise demonstrated by PP foams in theoretical research and various application areas, a disparity persists between the research and application of PP foaming technology and practical requirements. Researchers have successfully developed numerous high-performance PP materials applicable in automotive, flame retardant, electromagnetic shielding, and oil–water separation. Nevertheless, certain challenges must be addressed to facilitate their large-scale application in practical production. These challenges encompass difficulties in achieving mass production of high-performance PP, reducing technology costs, and the absence of a fully established recycling mechanism. Effectively addressing these challenges will pave a more feasible path for the practical application of PP foaming technology, facilitating its broader utilization across various fields.

Author contribution Gang Wang: wrote the main text, conceived and designed the review, editing. Mengyao Dong: Conceived and designed the review, editing. Huahong Deng: Formal analysis and editing. Xiao Ma: Data collection and curation. Bingyue Zhu: Formal analysis and editing. Li Zhou: Data collection and curation. Xiangning Zhang: Formal analysis and editing. Daqing Tan: Data collection and curation. Hassan Algadi: Review and editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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